

BRIAN SU

Incoming Embedded Engineer @ Nokia

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Education

University of Waterloo

Sep 2024 – May 2029

Honours Computer Science, Digital Hardware Specialization

Waterloo, ON

- GPA: 3.7/4.0
- Relevant Coursework: Computer Architecture, ASIC Systems, OOP

Technical Skills

Languages: C, C++, Python, Go, RISC-V Assembly, Java, SQL

Hardware: FPGA, VHDL, ASIC Design

Systems: TCP/IP, I2C, SPI, RTOS, Linux, Bash, Wireshark, Git

ML/Compute: CUDA, NumPy, TensorFlow, Keras, LoRA

Experience

Embedded Software Developer

Jan 2026 - present

Nokia

Ottawa, ON

- Designed a debug and analysis tool to detect high-cost rule updates with less than **0.01%** runtime overhead and negligible memory impact, for safe live configuration updates in latency-critical data paths
- Designed a data path optimization improving **throughput by 10%** in a hot path without additional baseline overhead
- Re-designed driver abstractions to allow for flexible changes to packet headers
- Integrated and tested datapath components on internal RTOS platforms

Software Developer

May 2025 - Oct 2025

Airfairness

Waterloo, ON

- Designed and scaled a database serving **1,000+** monthly users by implementing efficient data normalization and well-structured schemas, ensuring reliable and performant data access under increasing load
- Improved data retrieval speed by implementing a **caching layer**, retry backoff for API calls, and **IP-based rate limiting**, reducing redundant requests and cutting **latency by 42%**
- Performed an **end-to-end migration** of a database with **10,000 records**, enforcing strict type validation and normalization standards while optimizing storage and query performance
- Created routines with **Google Pub/Sub** and **cloud agents** with threading for optimized gmail parsing

Firmware Developer

Sep 2024 - May 2025

Waterloo Aerial Robotics Group

Markham, ON

- Effectively extracted Potentiometer information to a motor, utilizing the Hal library in **C** for signal conversion **APIs**.
- Programmed a **NucleoSTM32** processor to convert ADC signals to PWM using SPI communication.
- Performed in depth research on the **MCP3004/8**, SPI, I2C and UART for efficient data transmission

Projects

🔗 SLiM | SLM generation, training and hyper-specialization software

- Built a training software with **NVIDIA CUDA** to automatically fine tune base SLM models for hyperspecialized, customization use, shipping models retaining a **92% baseline accuracy** from the LLM.
- Created a real time LLM and memory routing system to create and split **fine tuning** datasets.

🔗 NanoWorks | AI DNA splicing and origami

- Developed a predictive **path finding algorithm** using Python and **NumPy** to determine optimal scaffolding positions in single-stranded DNA (ssDNA) structures.
- Engineered a DNA **behavior modeling** algorithm by extracting and processing structural data from filtered AI-generated imagery.

FlowScan | AI LiDAR to CFD simulator

- Developed an AI mesh filler in **CUDA**, and lightweight algorithm to fill gaps in LiDAR 3D image scans
- building an export pipeline to convert captured LiDAR meshes into 3D models for use in Computational Fluid Dynamics (**CFD**) simulations, enabling real-world geometry analysis.